EXPLOITS UNIVERSITY

BSc INFORMATION COMMUNICATION AND TECHNOLOGY

BICT 3202 · OBJECT-ORIENTED ANALYSIS AND DESIGN

WEEK 16

Comprehensive Revision, Integrated Case Study and Examination Preparation

PROGRAMME	BSc Information Communication and Technology
COURSE CODE / YEAR	BICT 3202 · Year 3
LECTURER	Francis Fweta — ICT Lecturer
DELIVERY	2h Lecture · 1h Tutorial · 1h Practical · 10h Independent Learning

Prepared by Francis Fweta · Exploits University · Week 16 of 16

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1. Introduction to Week 16

Welcome to the final teaching week of Object-Oriented Analysis and Design. This week is deliberately different: we are not introducing another isolated topic. Week 16 answers the most important question: *can the student take everything learned throughout the module and solve a new object-oriented analysis and design problem independently?*

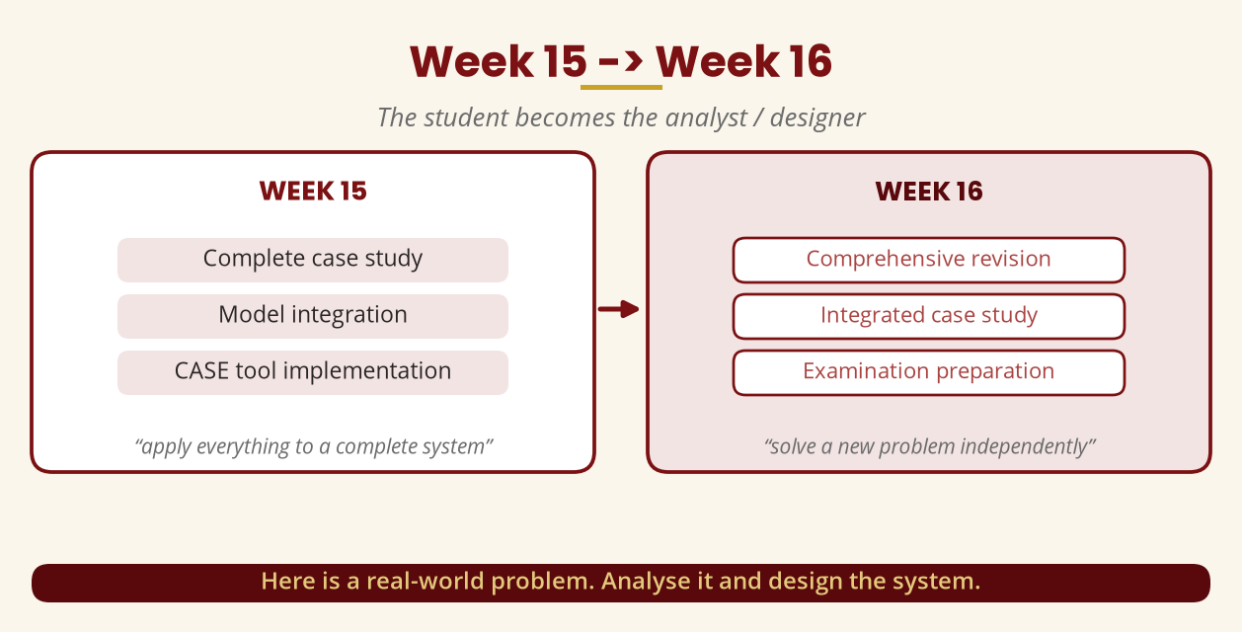


Figure 1 — Week 15 -> Week 16

2. Week 16 Learning Outcomes

By the end of this week, a student should be able to:

- Explain the principles of OOAD and the strengths and limitations of object-oriented development.
- Distinguish analysis from design; explain the purpose of UML and its diagram families.
- Identify actors, use cases, objects and classes from a problem statement.
- Construct use-case, class, state, activity, sequence and communication diagrams.
- Explain association, aggregation, composition, generalisation, inheritance and polymorphism.
- Explain the Unified Software Development Process — phases, iterations, workflows, artefacts.
- Combine object, dynamic and behavioural models; evaluate model consistency.
- Apply OOAD to an unfamiliar problem and answer examination questions effectively.

3. The Entire Module in One Picture

Students should be able to reproduce this conceptual journey from memory: understand the problem -> business needs -> requirements -> use cases -> identify objects/classes -> object model, dynamic model and behaviour model (classes, states, interactions) -> analysis model -> object-oriented design -> architecture/components -> implementation. **These are connected stages and views, not unrelated examination topics.**

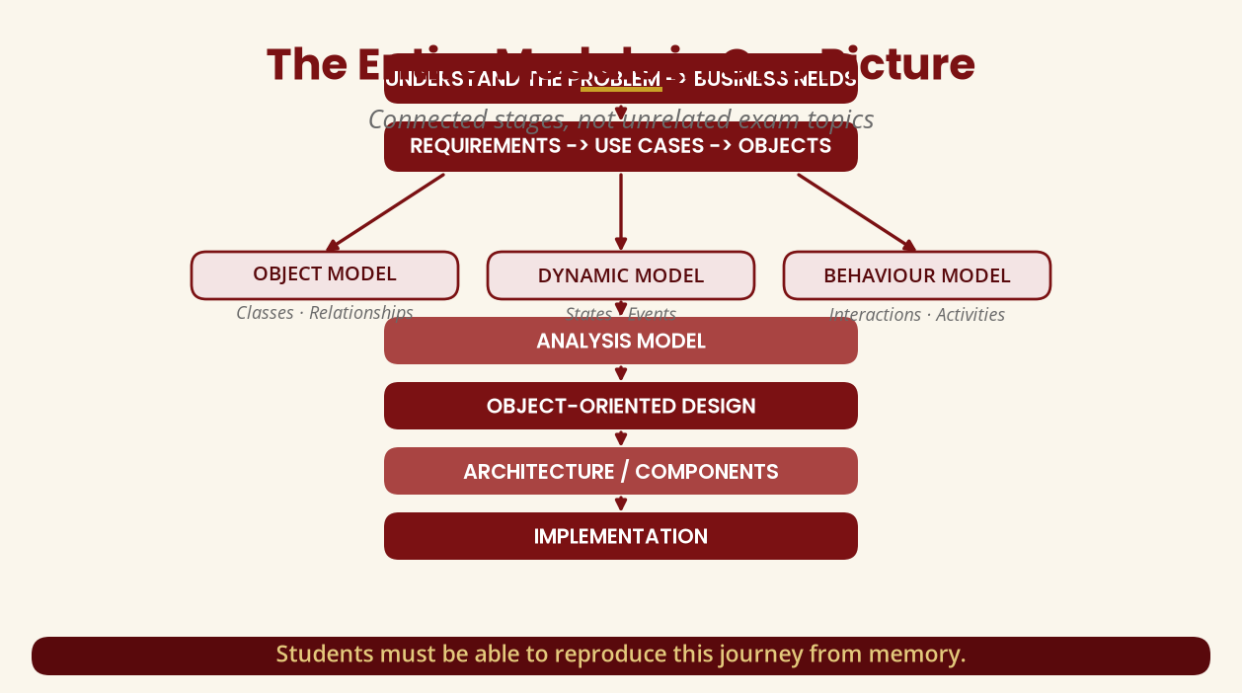


Figure 2 — The entire module in one picture

PART A — WHAT IS OOAD?

4. Object-Oriented Analysis and Design

Object-oriented analysis asks: *what does the system need to do, and what objects/classes exist in the problem domain?* For a library system the analyst may identify Student, Book, Librarian, Loan and Library, then investigate what a student does, what a book is, how a book is borrowed and returned, who may borrow, and the relationships between students and loans.

Object-oriented design asks: *how should the software be structured so those requirements can actually be implemented?* Where analysis identifies Student, Book and Loan, design may introduce LoanService, BookRepository, StudentRepository, NotificationService and LoanController. Therefore: **analysis focuses on understanding the problem; design focuses on constructing a software solution.**

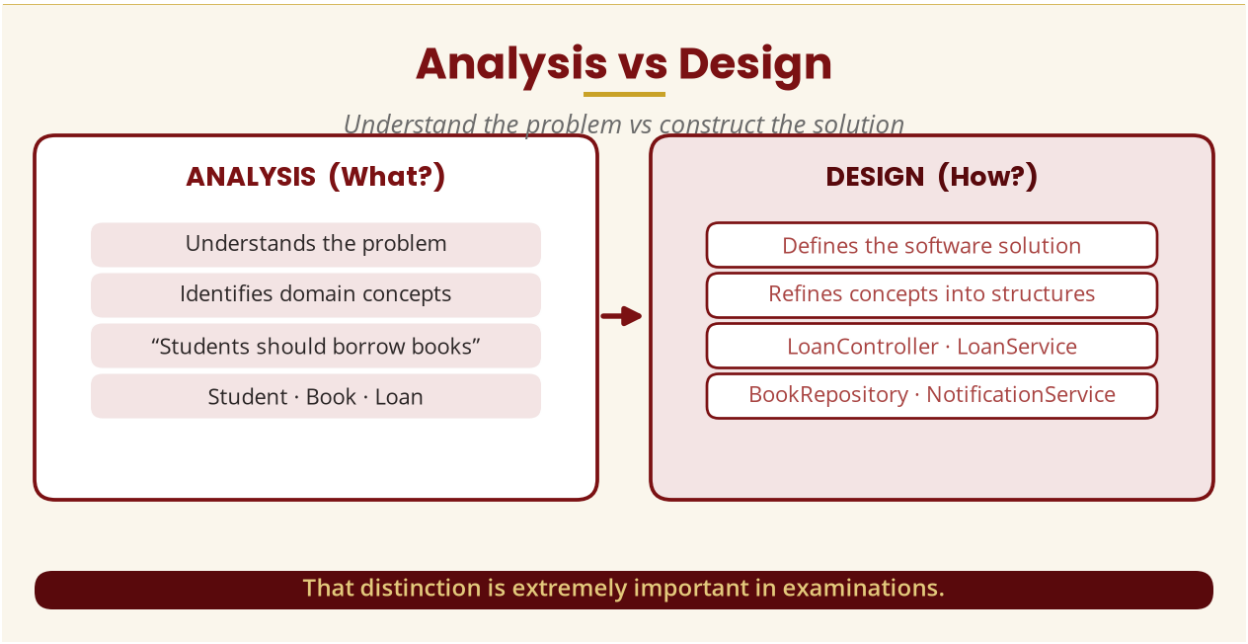


Figure 3 — Analysis vs design

Analysis	Design
Understands the problem	Defines the solution
What does the system need?	How will the system do it?
Identifies domain concepts	Refines them into software structures
Focuses on requirements	Focuses on implementation structure
"What?"	"How?"

PART B — OBJECT-ORIENTED PRINCIPLES REVISION

5. Object, Class, Encapsulation and Abstraction

An **object** represents a particular entity with **identity, state and behaviour** — Student #ST1001 with name = John, programme = ICT, year = 3 and behaviour registerCourse(), dropCourse(), viewResults(). A **class** is the blueprint from which objects are created; the object is an instance of that blueprint.

Encapsulation keeps an object's data and the operations that control that data together while controlling inappropriate direct access — a customer should not set `balance = -500000` directly; the BankAccount controls how balance changes through deposit(), withdraw() and getBalance(). **Abstraction** focuses on important characteristics while hiding unnecessary detail — an ATM shows Withdraw, Deposit and Check Balance without exposing database queries, encryption or server communication.

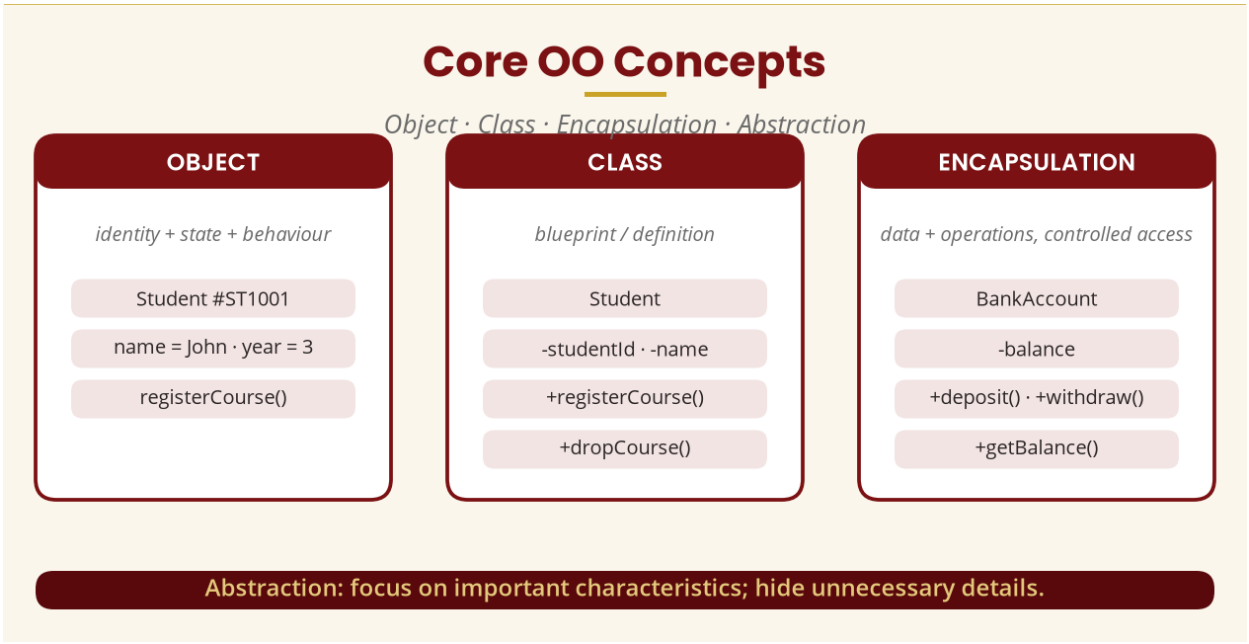


Figure 4 — Core object-oriented concepts

6. Inheritance and Polymorphism

Inheritance lets a specialised class inherit characteristics from a more general class — Student and Lecturer inherit name, email and login() from Person. **Polymorphism** means the same operation/interface can behave differently depending on the object involved — PermanentEmployee and ContractEmployee each implement `calculateSalary()` differently, so `employee.calculateSalary()` behaves differently per object.

Examination distinction: generalisation is the UML modelling relationship; inheritance is the object-oriented programming mechanism commonly used to implement it.

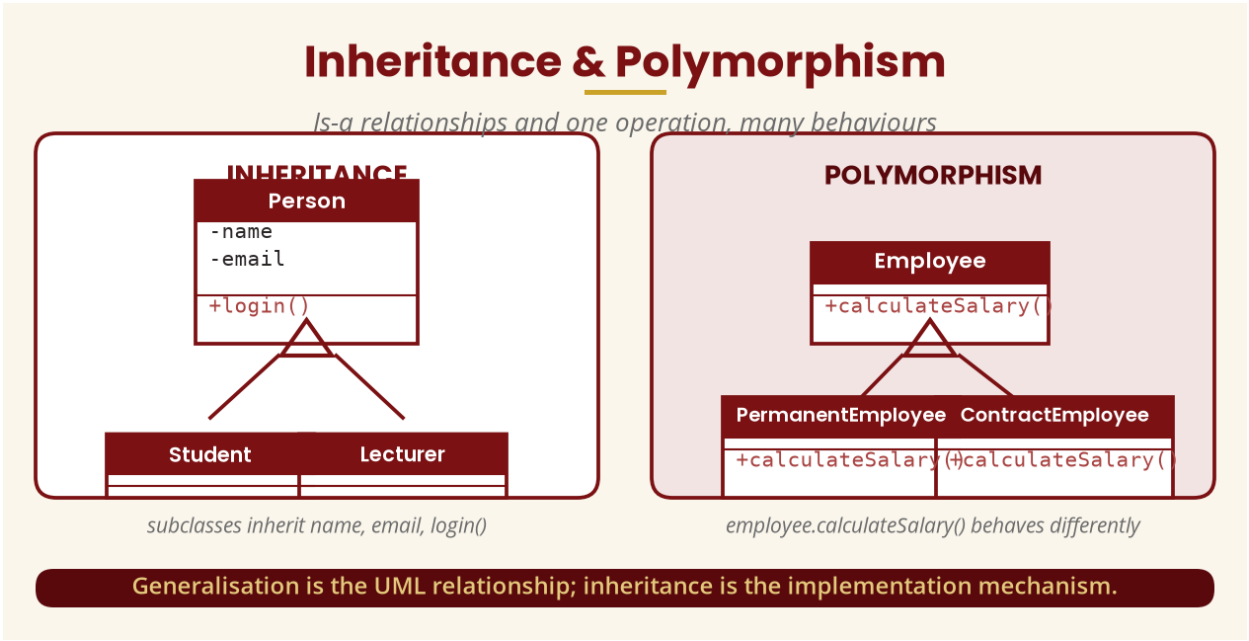


Figure 5 — Inheritance and polymorphism

PART C — UML REVISION

7. What Is UML, and Why Do We Need It?

UML is the Unified Modeling Language — a standard modelling language used to represent and communicate software/system models. The Object Management Group (OMG) lists UML 2.5.1 as a formal specification (adopted December 2017). Remember: **UML is a modelling language, not a programming language.** A complicated banking transaction described in a paragraph is hard to visualise; a diagram makes relationships and behaviour much easier to communicate.

8. UML Diagram Families

Structural diagrams show what the system is made of — class, component, deployment and package diagrams. **Behavioural diagrams** show what the system does — use-case, activity and state-machine diagrams. **Interaction diagrams** focus on communication between participants — sequence and communication/collaboration diagrams.

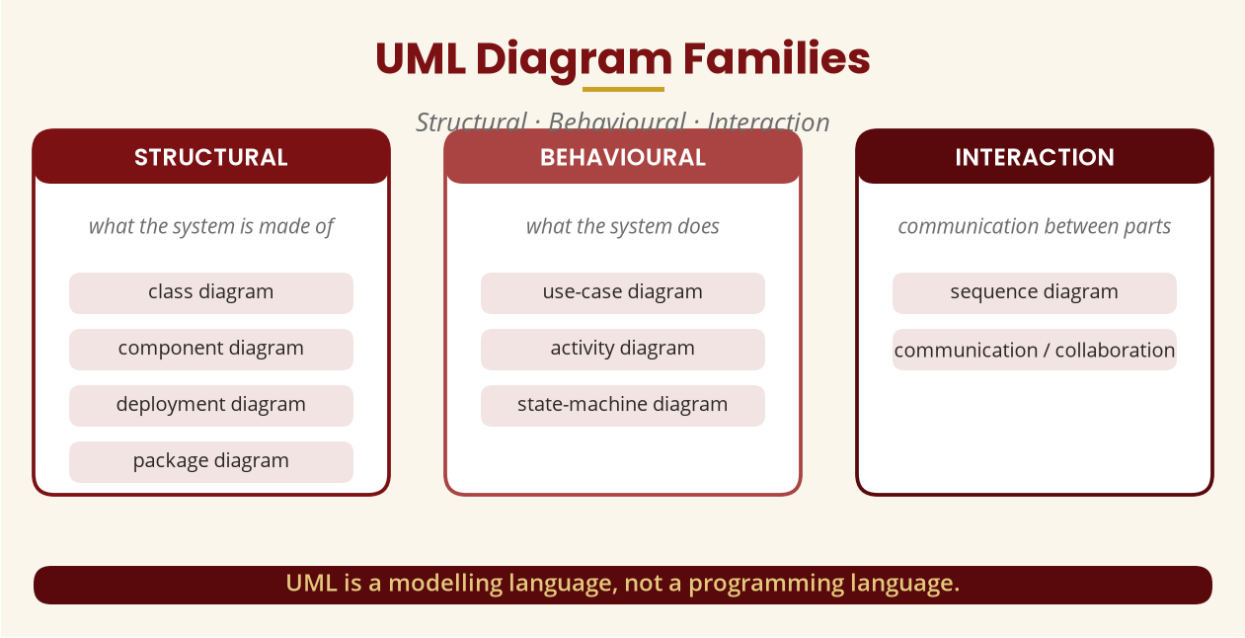


Figure 6 — UML diagram families

PART D — USE-CASE REVISION

9. Actor, Use Case, Include and Extend

A **use case** represents a goal an actor wants to accomplish with the system. The **actor** is the external role (Student); the **use case** is the goal (Register Course). `<<include>>` is used when one use case always incorporates required behaviour (Register Course includes Check Eligibility); `<<extend>>` is used for optional, conditional behaviour (Make Payment extended by Apply Discount). **Do not use include and extend randomly.**

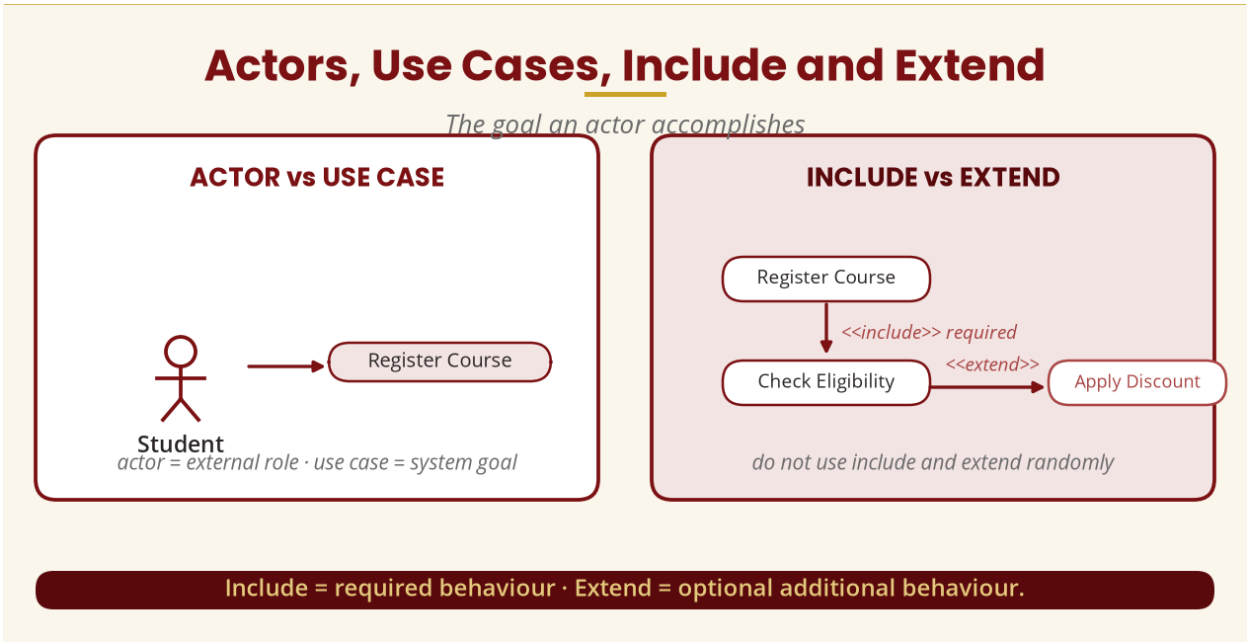


Figure 7 — Actors, use cases, include and extend

PART E — CLASS DIAGRAM REVISION

10. Class, Attribute, Operation and Visibility

The **class diagram** describes the static structure. An **attribute** is data belonging to a class (studentId, name, email); an **operation** is behaviour or responsibility (registerCourse(), dropCourse()). Common **visibility** symbols: + public, - private, # protected, ~ package — e.g. `-studentId : String` and `+ getName() : String`.

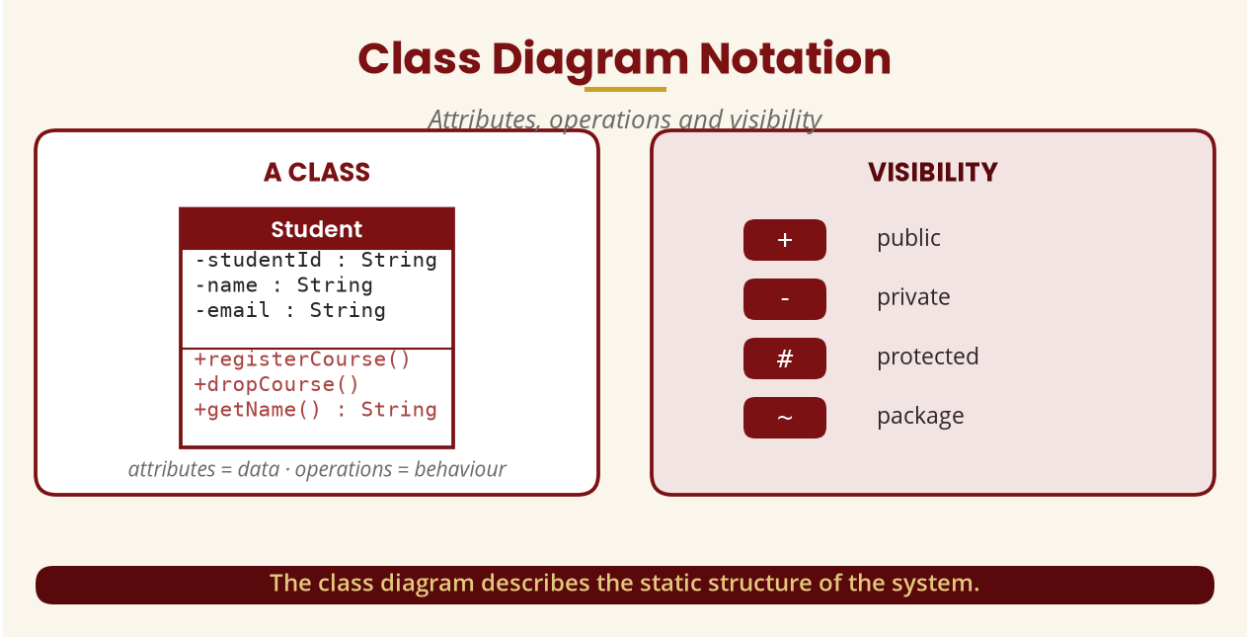


Figure 8 — Class diagram notation

PART F — RELATIONSHIPS REVISION

11. Association, Multiplicity, Aggregation, Composition, Generalisation

Association is a general relationship between classes (Student — Course). **Multiplicity** says how many objects participate: `1` exactly one, `0..1` zero or one, `*` many, `0..*` zero or many, `1..*` one or many — e.g. `Student 1 — 0..* Registration`.

Aggregation is a weaker whole-part relationship where parts can exist independently (Department o— Lecturer); **composition** is stronger ownership/lifecycle (House *— Room). **Generalisation** is an "is-a" relationship (Car and Motorcycle are Vehicles). Choose the diamond from the ownership/lifecycle relationship, not because "the diamond looks nice."

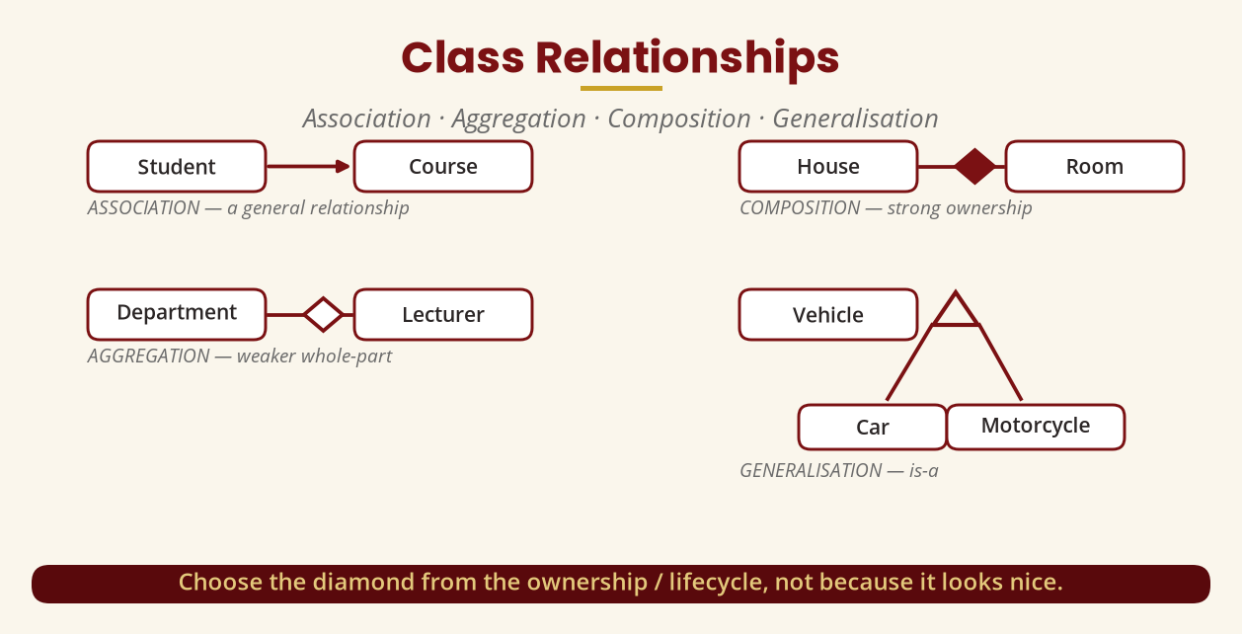


Figure 9 — Class relationships

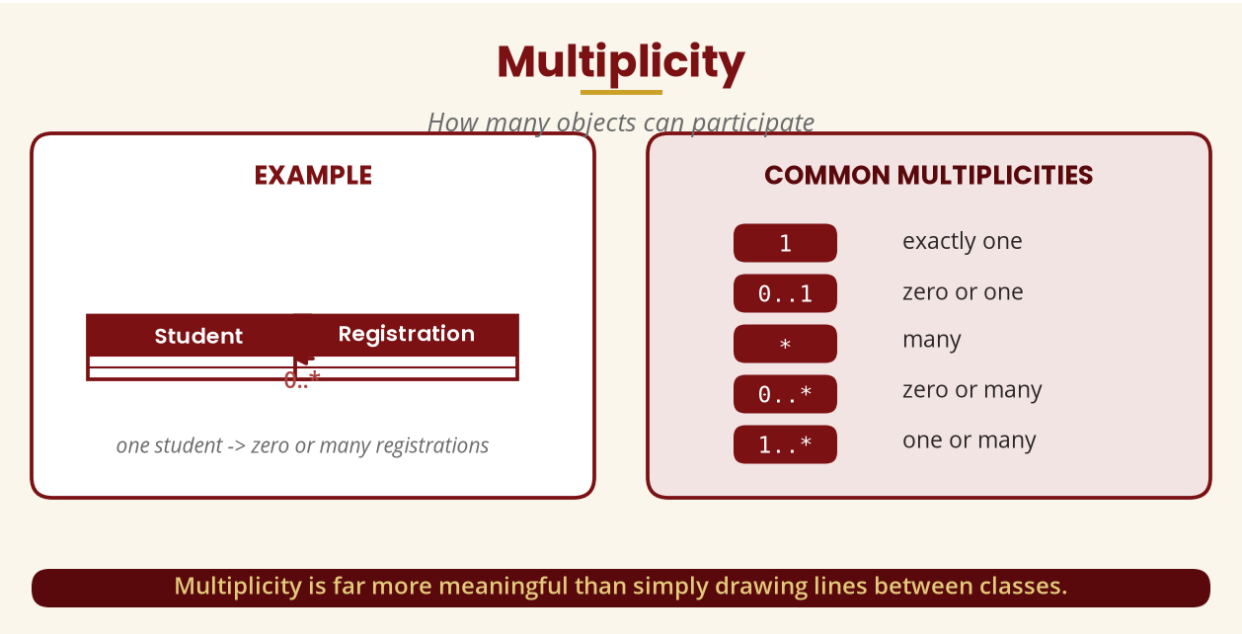


Figure 10 — Multiplicity

PART G — DYNAMIC MODELLING REVISION

12. Event, State, Transition and the State Diagram

Dynamic modelling describes how the system changes over time — *"what happens?"* rather than *"what exists?"*. An **event** causes a change (Course Registration Opens); a **state** is a condition at a point (Open, Closed, Full); a **transition** is movement between states (Available -> Full when the final seat is taken). The Course lifecycle: Created -> Available -> Full -> Closed.

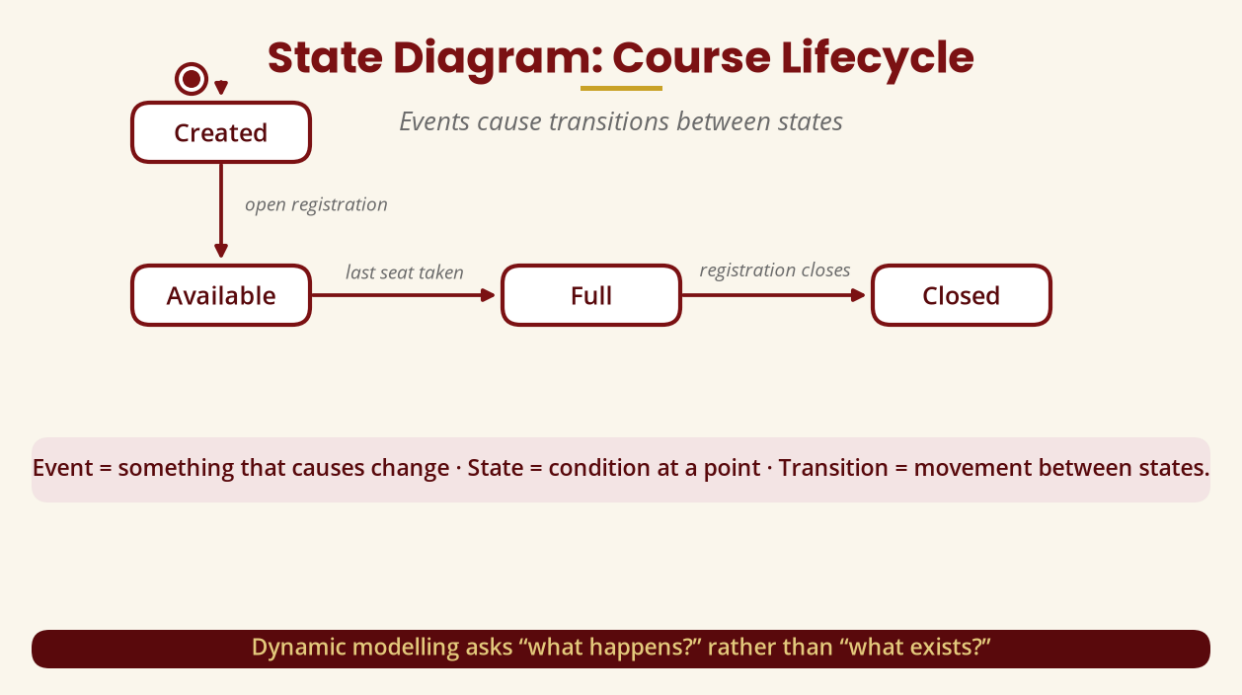


Figure 11 — State diagram: Course lifecycle

PART H — BEHAVIOURAL MODELLING REVISION

13. Activity, Sequence and Communication Diagrams

An **activity diagram** models a workflow (Start -> Login -> Select Course -> Check Eligibility -> Check Capacity -> Register -> End). A **sequence diagram** focuses on *who communicates with whom and in what order* (Student -> RegistrationController -> RegistrationService -> StudentRecord -> Registration). A **communication/collaboration diagram** also models interactions but emphasises objects, relationships and message numbering (1: register(), 2: validate(), 3: save()).

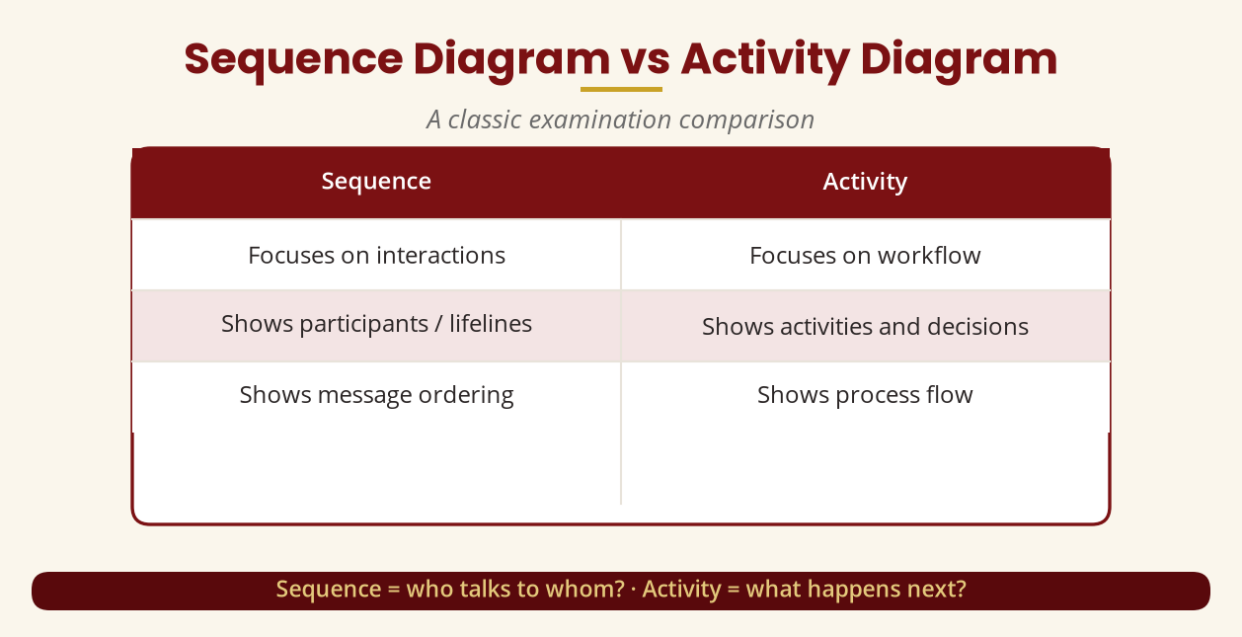


Figure 12 — Sequence diagram vs activity diagram

Sequence Diagram	Activity Diagram
Focuses on interactions	Focuses on workflow
Shows participants/lifelines	Shows activities/actions
Shows message ordering	Shows process flow

PART I — FULL MODEL INTEGRATION

14. The Most Important Week 16 Concept

Follow one requirement through the chain: **Requirement** ("students must be able to register for courses") -> **Use Case** (Register Course) -> **Classes** (Student, Course, Registration, RegistrationService) -> **Sequence** (Student -> RegistrationService -> Course -> Registration) -> **State** (Course: Available -> Full) -> **Activity** (Select Course -> Check Eligibility -> Check Capacity -> Register) -> **Design** (Presentation -> Application -> Domain -> Persistence). That is integrated OOAD.

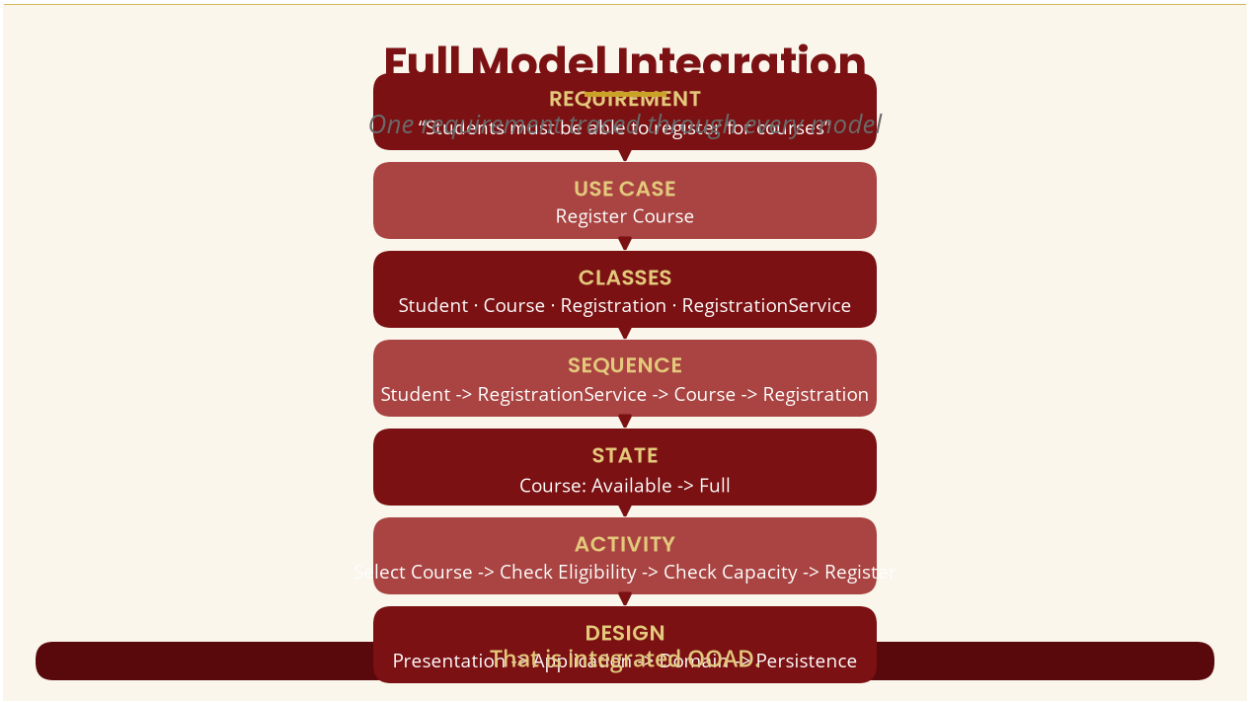


Figure 13 — Full model integration

PART J — UML MODEL CONSISTENCY

15. What Does Consistency Mean?

If the class diagram says `Course.checkCapacity()` but the sequence diagram says `Course.checkAvailableSeats()`, ask: *are these the same operation?* If not, the model may be inconsistent. Perform the six checks: (1) use cases <- requirements; (2) use cases <- classes; (3) sequence <- class; (4) activity <- sequence; (5) state <- behaviour; (6) design <- requirements.

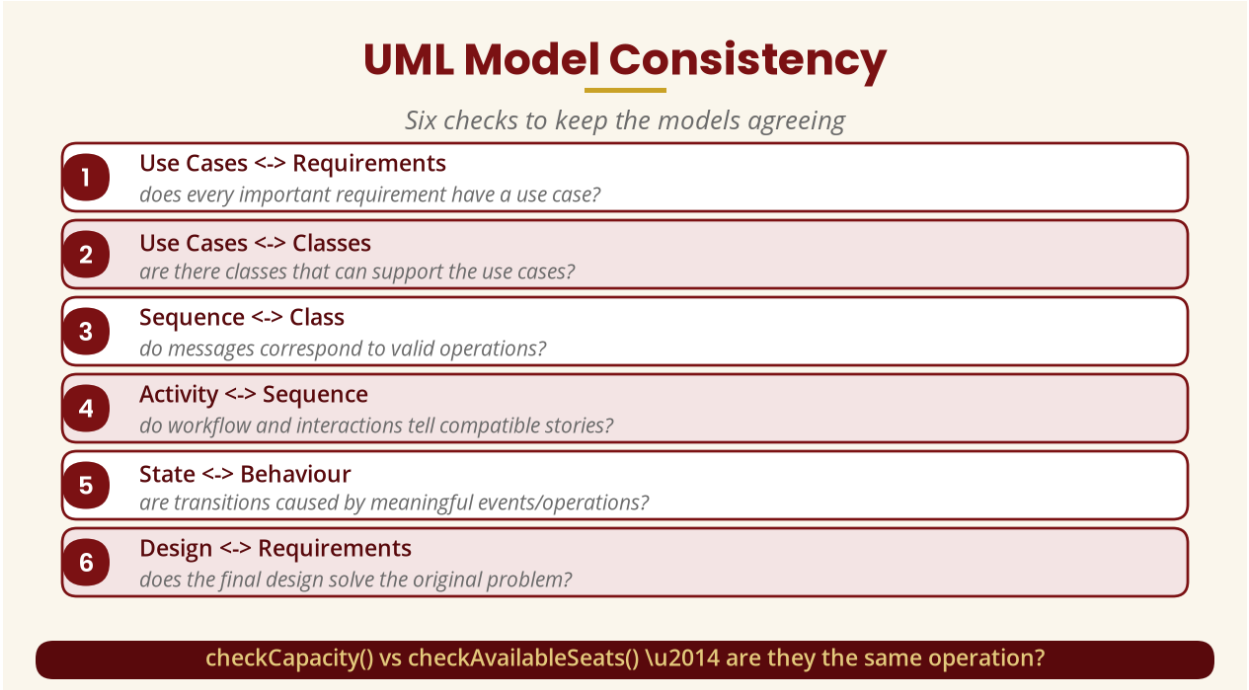


Figure 14 — Six model-consistency checks

PART K — UNIFIED SOFTWARE DEVELOPMENT PROCESS REVISION

16. USDP/UP and Its Four Phases

The Unified Software Development Process (USDP), associated with the Unified Process, is an **iterative and incremental** approach: development is divided into phases and iterations rather than one uninterrupted sequence. IBM's Rational Unified Process documentation describes four phases:

Inception — "should we build this system?" (business problem, scope, stakeholders, feasibility, business case). **Elaboration** — "do we understand the problem and architecture?" (requirements, architecture, risk, key use cases, initial design). **Construction** — "can we build it?" (implementation, integration, testing, refinement). **Transition** — "can users use it?" (deployment, acceptance, training, fixes, operational use).

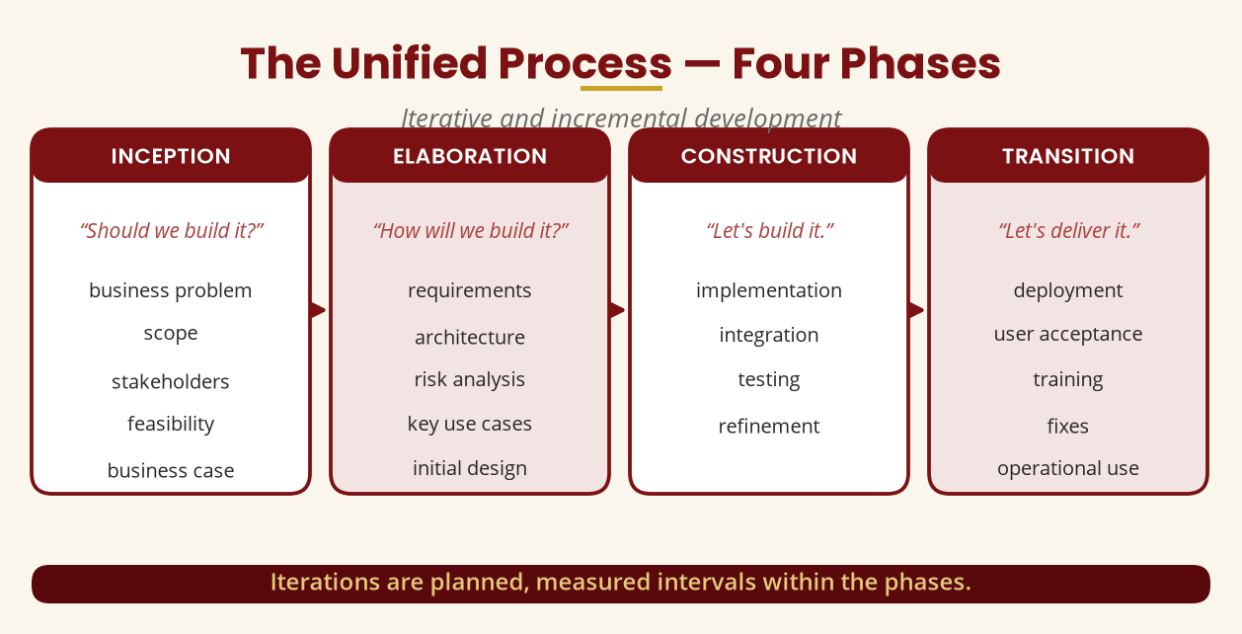


Figure 15 — The four phases of the Unified Process

PART L — ITERATIVE DEVELOPMENT

17. What Is an Iteration?

An **iteration** is a development cycle during which a team produces and evaluates part of the system. Instead of "plan everything -> build everything -> test everything", an iterative approach cycles through: Iteration 1 (requirements + basic design) -> feedback -> Iteration 2 (more functionality) -> feedback -> Iteration 3 (more functionality) -> feedback -> Iteration 4 (refinement + testing). Iterations are planned, measured intervals within phases that deliver incremental value to stakeholders.

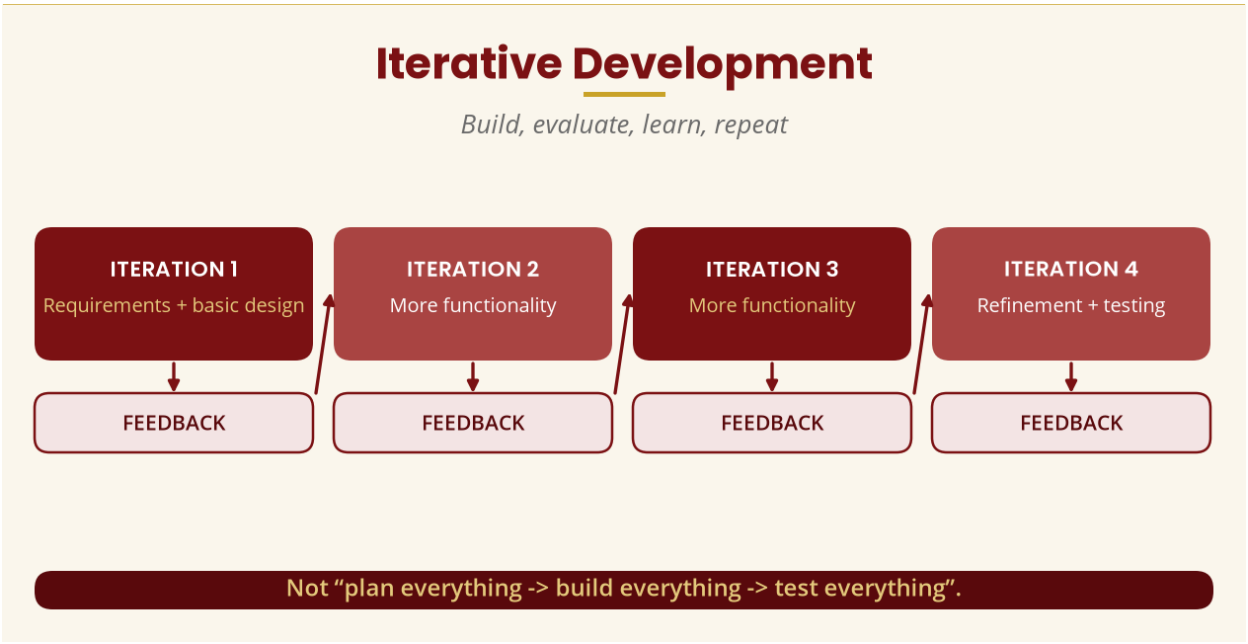


Figure 16 — Iterative development

PART M — REUSE REVISION

18. What Is Software Reuse?

Software reuse means using an existing component, class, library, framework, service, design or architecture rather than developing everything from scratch — an authentication library instead of a bespoke mechanism, or a UI framework instead of building every control. The goal: reuse -> less duplication -> less effort -> potentially improved consistency -> easier maintenance. However, reuse should be based on suitability, security, maintainability, compatibility and licensing — not simply because something already exists.

PART N — CASE STUDY FOR FINAL REVISION

19. Scenario: Mobile Money System

A telecommunications company wants a mobile money system. **Users** create accounts, deposit, withdraw, send, receive, check balance and view history. **Agents** deposit and withdraw for customers and view their transactions. **Administrators** manage users and agents, monitor transactions and generate reports.

Actors: Customer, Agent, Administrator, Banking System, Notification Service. **Candidate classes:** User, Customer, Agent, Administrator, Wallet, Transaction, Deposit, Withdrawal, Transfer, Notification — plus services AuthenticationService, TransactionService, NotificationService. **Relationships:** User generalises Customer / Agent / Administrator; `Customer 1 — 1 Wallet`; `Wallet 1 — 0..* Transaction`. **Send Money:** sendMoney() -> validateSender() -> checkBalance() -> creditReceiver() -> createTransaction() -> sendNotification(). **Transaction states:** Created -> Pending -> Completed (or Failed).

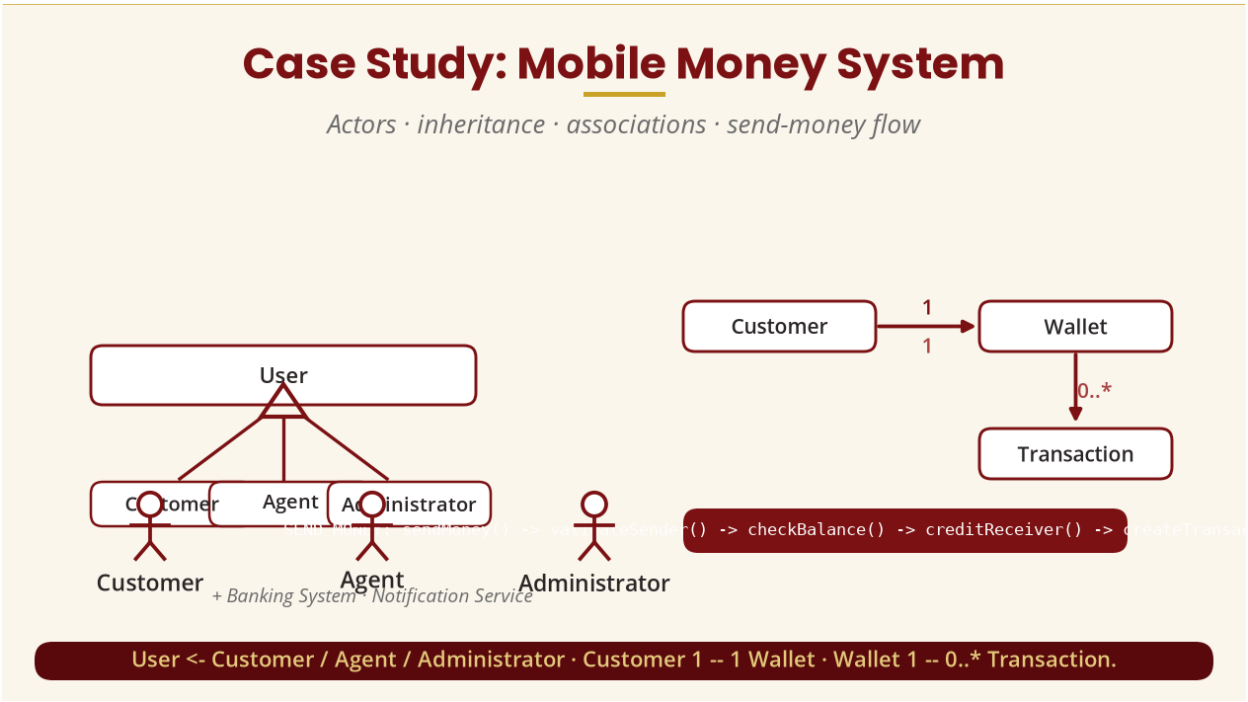


Figure 17 — Mobile money case study

PART O — PRACTICAL REVISION

20. Practical Exercise and Challenge

Use a UML/CASE tool to model the mobile money system with six diagrams: use-case, class, sequence (Send Money), activity (Send Money), state-machine (Transaction) and component. Then deliberately introduce the error: the sequence diagram calls `validateTransaction()` but the class diagram contains `checkTransaction()` — and correct it. **UML modelling is complete when the model is logically consistent, not when the diagram looks beautiful.**

PART P — EXAMINATION REVISION

21. Common Theory and Practical Questions

Theory: define OOAD; explain five characteristics of OO systems; distinguish an object from a class; explain encapsulation, abstraction, inheritance and polymorphism; what is UML; structural vs behavioural modelling; association vs aggregation vs composition; generalisation vs inheritance; purpose of a use-case diagram; sequence vs activity diagrams.

Practical: given a scenario — identify actors and use cases, draw the use-case diagram, identify classes, draw the class diagram with multiplicities and inheritance, create sequence, activity, state and component diagrams, and explain how the diagrams relate.

PART Q — HOW TO APPROACH AN EXAM SCENARIO

22. Never Start Drawing Immediately

Follow a disciplined process: (1) read the problem twice; (2) underline the nouns; (3) identify possible objects/classes; (4) identify actors; (5) identify actions/goals; (6) convert goals into use cases; (7) build the use-case diagram; (8) develop the class model; (9) choose an important scenario; (10) develop the sequence diagram; (11) develop activity/state models; (12) check consistency.

The noun technique: in "a patient books an appointment with a doctor; the receptionist manages patient records and doctors access their schedules", candidate nouns are Patient, Appointment, Doctor, Receptionist, Patient Record and Schedule; candidate verbs are book, manage and access. This is a useful candidate-identification technique — not a mechanical rule that every noun must become a class.

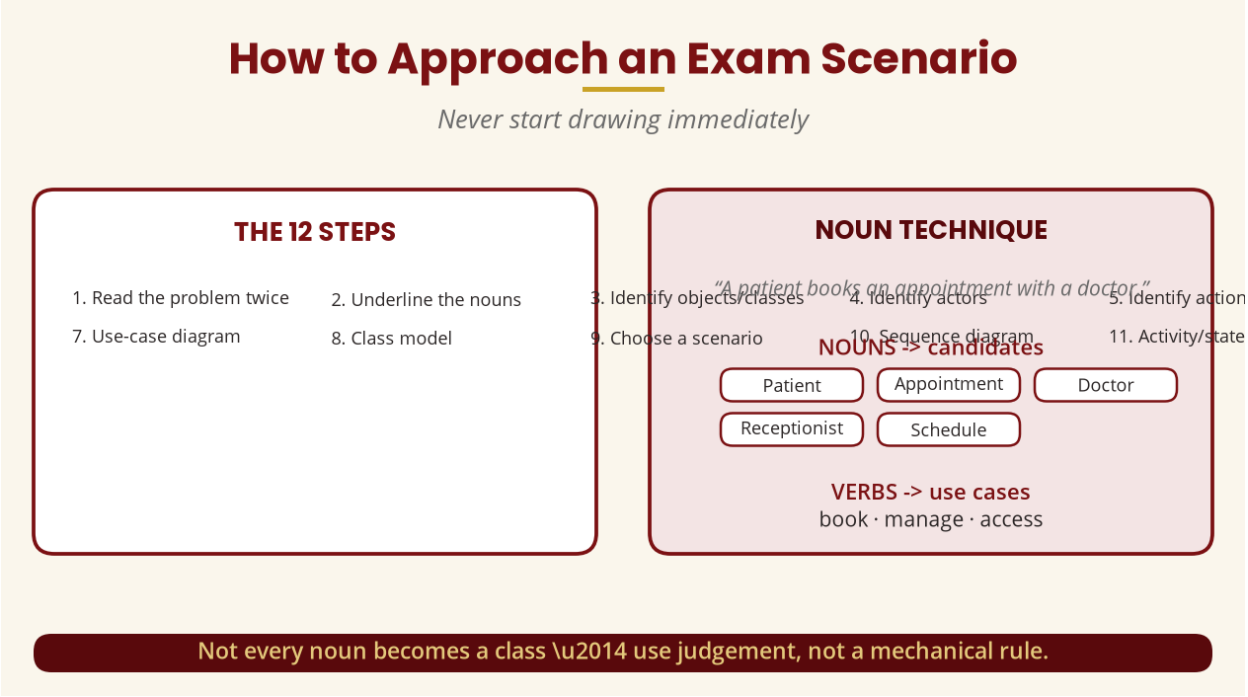


Figure 18 — How to approach an exam scenario

PART R — COMMON STUDENT MISTAKES

- **Confusing class and object** — Student can be a class; student123 is an object/instance.
- **Every noun becomes a class** — "quickly" is not a class; use judgement.
- **Every verb becomes a method** — ask which object has the responsibility.
- **Confusing include and extend** — include = required; extend = optional.
- **Confusing aggregation and composition** — decide from ownership/lifecycle.
- **Sequence diagrams without class diagrams** — messages need a home in the class model.
- **Beautiful UML, wrong logic** — examiners mark correct modelling, not decoration.

PART S — MASTER REVISION TABLE

23. All Major Concepts

Topic	Main Question
Business needs	Why does the organisation need the system?
Requirements	What does the system need to provide?
Object / Class	What entity exists? / What defines similar objects?
Encapsulation / Abstraction	How are data and behaviour controlled? / What details are hidden?
Inheritance / Polymorphism	What can a subclass inherit? / How can one operation behave differently?
UML	How do we communicate the model visually?
Use case / Class diagram	What goal does an actor accomplish? / What structure exists?
Association / Aggregation / Composition	How are classes related? / What whole-part relationship exists?
Generalisation	What is the superclass/subclass relationship?
Dynamic model / State / Event	How does the system change over time?
Activity / Sequence / Communication	What is the workflow? Who communicates, and when?
USDP / Reuse / Design / Component	How is development organised? What can be reused? How is the solution structured?
CASE tool / Integration	How is modelling supported by software? Do all models tell the same story?

PART T — THE 16-WEEK JOURNEY

24. What Students Have Covered

- **Week 1:** business needs, software needs and object technology for business.
- **Week 2:** principles of object modelling.
- **Weeks 3–4:** UML overview, building blocks, rules, mechanisms and architecture.
- **Weeks 5–6:** object modelling I & II — classes, associations, aggregation, inheritance, polymorphism and grouping.
- **Week 7:** dynamic modelling — events, states, operations and concurrency.

- **Week 8:** behavioural modelling — use cases, interactions, sequences, communication and activities.
- **Week 9:** USDP/UP — phases, iterations, workflows, artefacts, risk and reuse.
- **Weeks 10–12:** object-oriented design, refinement and integrated modelling.
- **Weeks 13–14:** multilayer architecture, reusable components and integrated design.
- **Week 15:** complete OOAD case study and CASE-tool implementation.
- **Week 16:** comprehensive revision, integrated application and examination preparation.

25. Tutorial, Practical and Examination Practice

Tutorial: an Online Examination Management System — identify actors; at least 10 use cases; at least 10 candidate classes; the use-case diagram; the class diagram; a sequence diagram for Start Examination; an activity diagram for Taking an Examination; a state diagram for Examination (Created -> Scheduled -> Open -> In Progress -> Submitted -> Marked -> Released); identify components; explain how the diagrams relate.

Practical mini-project: choose a Hospital, Banking, Mobile Money, University Registration, Library, Hotel, E-commerce or Transport system, and submit: problem statement, objectives, functional and non-functional requirements, actors, use cases, use-case, class, sequence, activity, state and component diagrams, architecture, model consistency analysis and conclusion.

Revision question bank: use the short-answer, comparison, long-answer and case-study prompts in this module for untimed practice; they are not a complete timed paper. Two parallel full-length practice papers are provided as `exams/End-Semester-Exam-Paper-A.md` and `exams/End-Semester-Exam-Paper-B.md`, with a separate confidential marking guide. Each draft paper is 100 marks with Q1 compulsory (40) plus any three of four 20-mark questions. The course outline assigns 60% to the examination but does not specify a duration; the three-hour duration in the drafts must be confirmed by the Faculty before formal issue.

26. Exam Answering Strategy

For **definition** questions, write the full definition and give an example. For **explain** questions, use Definition -> Explanation -> Example -> Importance. For **compare** questions, use a table. For **UML** questions, identify the requirement, actors, use cases, classes, relationships and behaviour — then check consistency. State assumptions, and let the marks guide the amount of detail. **Do not randomly draw diagrams.**

For the proposed 3-hour practice papers, a possible plan is about 65 minutes for the 40-mark integrated case, about 32 minutes for each of the three chosen Section B questions, and the remaining time for reading and checking. Adjust this plan if the approved exam duration differs.

27. Final Week 16 Takeaway

THE ENTIRE MODULE IN ONE QUESTION

How can we systematically understand a software problem and represent a solution using objects, classes, UML models and an appropriate development process? The answer: **Business Problem -> Requirements -> Actors -> Use Cases -> Objects/Classes -> Class Model -> Dynamic/State Model -> Behavioural/Interaction -> Analysis -> Design -> Architecture/Components -> Model Integration -> Implementation.** OOAD is not simply about drawing UML diagrams — it is about thinking systematically, identifying what matters, understanding interactions and behaviour, and transforming that understanding into a coherent design developers can implement.

28. Final Revision Checklist

- **Object orientation:** can I define object, class, encapsulation, abstraction, inheritance, polymorphism?
- **UML:** can I explain UML and distinguish structural, behavioural and interaction diagrams?
- **Use cases:** can I identify actors and use cases, and use include/extend correctly?
- **Class modelling:** can I identify classes, attributes, operations, associations, multiplicities, aggregation, composition and generalisation?
- **Dynamic & behavioural:** can I model events, states, transitions, activity, sequence and communication diagrams?
- **USDP:** can I explain Inception, Elaboration, Construction, Transition, iterations, workflows and artefacts?
- **Design:** can I distinguish analysis from design, combine the models, identify components, explain layered architecture and reuse?
- **Practical:** can I use a CASE tool, model a new system and check consistency?
- **Examination:** can I explain in my own words, give examples, draw diagrams and justify decisions?

29. References and Further Reading

Primary standards and process frameworks

- Object Management Group (OMG). *Unified Modeling Language (UML), Version 2.5.1.* OMG — formal UML specification and normative documents.
- IBM. *Rational Unified Process (RUP)* — four phases (Inception, Elaboration, Construction, Transition) with iterations as planned intervals.

Prescribed and recommended texts

- Mach, E. (2019). *Object Oriented Analysis & Design Cookbook: Introduction to Practical System Modeling.*

- Reddy, V., & Korra, S. (2018). *Object-Oriented Analysis and Design Using UML.*
- The Art of Service. (2021). *Object Oriented Analysis and Design: A Complete Guide.*
- Wixom, B., & Dennis, A. (2018). *Systems Analysis and Design.*
- Blokdyk, G. (2021). *Object-Oriented Analysis and Design Standard Requirements.*
- Wixom, B., & Dennis, A. (2020). *Systems Analysis and Design: An Object-Oriented Approach with UML.*

30. Conclusion of the 16-Week Module

Week 1 establishes business/software needs and object technology; Week 2 develops object modelling principles; Weeks 3–4 cover UML; Weeks 5–6 develop object modelling; Week 7 covers dynamic modelling; Week 8 covers behavioural modelling; Week 9 introduces USDP/UP; Weeks 10–12 develop object-oriented design, refinement and model integration; Weeks 13–14 focus on multilayer architecture, reusable components and design integration; Week 15 applies the process in a complete case study. In Week 16, **the student becomes the analyst/designer** — able to move from Requirements -> Actors -> Use Cases -> Classes -> Relationships -> Interactions -> Behaviour -> Architecture -> Components -> an Integrated UML Model. That is the culmination of BICT 3202 Object-Oriented Analysis and Design.